

**Assignment 5**

Semester I, 2025/26

1. [15 pts] Rotation Matrices

The matrix operators corresponding to rotations $R_x(\phi)$ and $R_y(\phi)$ through a counterclockwise angle ϕ about the x and y axes respectively are given by

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{pmatrix}, \quad R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & -\sin \phi \\ 0 & 1 & 0 \\ \sin \phi & 0 & \cos \phi \end{pmatrix}.$$

- (a) The matrix operator corresponding to a rotation through ϕ_1 about the x -axis, followed by a rotation through ϕ_2 about the y -axis is given by

$$R_{xy}(\phi_1, \phi_2) \equiv R_y(\phi_2)R_x(\phi_1).$$

Calculate $R_{xy}(\phi_1, \phi_2)$ and $R_{yx}(\phi_2, \phi_1)$. Do $R_x(\phi_1)$ and $R_y(\phi_2)$ commute?

- (b) Write an expression for the inverse matrix $R_{xy}^{-1}(\phi_1, \phi_2)$ in terms of $R_x(\phi)$ and $R_y(\phi)$ and hence verify explicitly that

$$R_{xy}^{-1}(\phi_1, \phi_2) = R_{xy}^T(\phi_1, \phi_2),$$

and show that $\det[R_{xy}^{-1}(\phi_1, \phi_2)] = 1$.

Answer:

(a)

$$\begin{aligned} R_{xy}(\phi_1, \phi_2) &= \begin{pmatrix} \cos \phi_2 & 0 & -\sin \phi_2 \\ 0 & 1 & 0 \\ \sin \phi_2 & 0 & \cos \phi_2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi_1 & \sin \phi_1 \\ 0 & -\sin \phi_1 & \cos \phi_1 \end{pmatrix} \\ &= \begin{pmatrix} \cos \phi_2 & \sin \phi_1 \sin \phi_2 & -\cos \phi_1 \sin \phi_2 \\ 0 & \cos \phi_1 & \sin \phi_1 \\ \sin \phi_2 & -\sin \phi_1 \cos \phi_2 & \cos \phi_1 \cos \phi_2 \end{pmatrix} \end{aligned} \quad (1)$$

and

$$\begin{aligned} R_{yx}(\phi_2, \phi_1) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi_1 & \sin \phi_1 \\ 0 & -\sin \phi_1 & \cos \phi_1 \end{pmatrix} \begin{pmatrix} \cos \phi_2 & 0 & -\sin \phi_2 \\ 0 & 1 & 0 \\ \sin \phi_2 & 0 & \cos \phi_2 \end{pmatrix} \\ &= \begin{pmatrix} \cos \phi_2 & 0 & -\sin \phi_2 \\ \sin \phi_1 \sin \phi_2 & \cos \phi_1 & \sin \phi_1 \cos \phi_2 \\ \cos \phi_1 \sin \phi_2 & -\sin \phi_1 & \cos \phi_1 \cos \phi_2 \end{pmatrix} \end{aligned} \quad (2)$$

Obviously, $R_{xy}(\phi_1, \phi_2)$ does not equal $R_{yx}(\phi_2, \phi_1)$ and $R_x(\phi_1)$ and $R_y(\phi_2)$ do not commute.

- (b) The matrix $R_{xy}^{-1}(\phi_1, \phi_2)$ can be interpreted as an operation that rotate opposite to the original direction $-\phi$. As a result:

$$R_{xy}^{-1}(\phi_1, \phi_2) = R_{yx}(-\phi_2, -\phi_1) = R_x(-\phi_1)R_y(-\phi_2) \quad (3)$$



write it down explicitly.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi_1 & -\sin \phi_1 \\ 0 & \sin \phi_1 & \cos \phi_1 \end{pmatrix} \begin{pmatrix} \cos \phi_2 & 0 & \sin \phi_2 \\ 0 & 1 & 0 \\ -\sin \phi_2 & 0 & \cos \phi_2 \end{pmatrix} = \begin{pmatrix} \cos \phi_2 & 0 & \sin \phi_2 \\ \sin \phi_1 \sin \phi_2 & \cos \phi_1 & -\sin \phi_1 \cos \phi_2 \\ -\cos \phi_1 \sin \phi_2 & \sin \phi_1 & \cos \phi_1 \cos \phi_2 \end{pmatrix} \quad (4)$$

obviously it is $R_{xy}^T(\phi_1, \phi_2)$. And the determinant can be derived trivially:

$$\det[R_{xy}^{-1}(\phi_1, \phi_2)] = \cos^2 \phi_1 \cos^2 \phi_2 + \sin^2 \phi_1 \cos^2 \phi_2 + \sin^2 \phi_1 \sin^2 \phi_2 + \cos^2 \phi_1 \sin^2 \phi_2 = 1 \quad (5)$$

2. [10 pts] Determinant Recurrence Relation

The $N \times N$ determinant Δ_N is given by

$$\Delta_N = \begin{vmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -2 & \end{vmatrix} ..$$

Establish a recurrence relation for $S_N \equiv \Delta_N + \Delta_{N-1}$ and hence find an explicit formula for Δ_N .

Answer:

Expand the determinant by the first row:

$$\begin{aligned} \Delta_N &= -2 \begin{vmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -2 & \end{vmatrix}_{(N-1) \times (N-1)} - \begin{vmatrix} 1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -2 & \end{vmatrix}_{(N-1) \times (N-1)} \\ &= -2\Delta_{N-1} - \begin{vmatrix} -2 & 1 & 0 & 0 & \cdots & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & -2 & \end{vmatrix}_{(N-2) \times (N-2)} = -2\Delta_{N-1} - \Delta_{N-2} \end{aligned} \quad (6)$$

Substitute (6) to $S_N = \Delta_N + \Delta_{N-1}$,

$$S_N = \Delta_N + \Delta_{N-1} = -\Delta_{N-1} - \Delta_{N-2} = -S_{N-1} = \dots = (-1)^{N-3} S_3 = (-1)^N \quad (7)$$

So the recurrence relation for determinants is

$$\Delta_N = (-1)^N - \Delta_{N-1} = (-1)^N - (-1)^{N-1} + \Delta_{N-2} = \dots = (-1)^N (N-2) + (-1)^{N-2} \Delta_2 \quad (8)$$

And we get the answer:

$$\Delta_N = (-1)^N (N+1) \quad (9)$$



3. [30 pts] Matrix Properties

Consider the following matrices:

$$A = \begin{pmatrix} 0 & -i & i \\ i & 0 & -i \\ -i & i & 0 \end{pmatrix}, \quad B = \frac{1}{\sqrt{8}} \begin{pmatrix} \sqrt{3} & -\sqrt{2} & -\sqrt{3} \\ 1 & \sqrt{6} & -1 \\ 2 & 0 & 2 \end{pmatrix}.$$

- Are they (i) symmetric, (ii) anti-symmetric, (iii) singular, (iv) Hermitian, (v) anti-Hermitian, (vi) unitary, or (vii) normal?
- If the matrix is neither symmetric nor anti-symmetric, decompose it into a sum of symmetric and anti-symmetric matrices.
- If the matrix is neither Hermitian nor anti-Hermitian, decompose it into a sum of Hermitian and anti-Hermitian matrices.
- Evaluate A^{20} .

Answer:

(a) For A,

$$A^T = \begin{pmatrix} 0 & i & -i \\ -i & 0 & i \\ i & -i & 0 \end{pmatrix} = -A, \quad A^\dagger = \begin{pmatrix} 0 & -i & i \\ i & 0 & -i \\ -i & i & 0 \end{pmatrix} = A, \quad AA^\dagger = A^\dagger A = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \quad (10)$$

As results, A is anti-symmetric, Hermitian and normal. For B,

$$B^T = B^\dagger = \frac{1}{\sqrt{8}} \begin{pmatrix} \sqrt{3} & 1 & 2 \\ 2\sqrt{2} & \sqrt{6} & 0 \\ -\sqrt{3} & -1 & 2 \end{pmatrix}, \quad BB^\dagger = B^\dagger B = \frac{1}{8} \begin{pmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} = I \quad (11)$$

As results, B is neither symmetric nor anti-symmetric and neither Hermitian nor anti-Hermitian, but it is normal and unitary.

(b, c) The symmetric and Hermitian part is:

$$\frac{1}{2}(B + B^T) = \frac{1}{2\sqrt{8}} \begin{pmatrix} 2\sqrt{3} & 1 - 2\sqrt{2} & 2 - \sqrt{3} \\ 1 - 2\sqrt{3} & 2\sqrt{6} & -1 \\ 2 - \sqrt{3} & -1 & 4 \end{pmatrix} = B_S \quad (12)$$

The anti-symmetric and anti-Hermitian part is:

$$\frac{1}{2}(B - B^T) = \frac{1}{2\sqrt{8}} \begin{pmatrix} 0 & 1 + \sqrt{2} & 2 + \sqrt{3} \\ -1 - \sqrt{2} & 0 & 1 \\ -2 - \sqrt{3} & -1 & 0 \end{pmatrix} = B_A \quad (13)$$

(d)

$$A^2 = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \quad A^3 = AA^2 = \begin{pmatrix} 0 & -3i & 3i \\ 3i & 0 & -3i \\ -3i & 3i & 0 \end{pmatrix} = 3A \quad (14)$$



As the result, when n is odd,

$$A^n = 3^{\frac{n-1}{2}} A \quad (15)$$

and when n is even,

$$A^n = 3^{\frac{n-2}{2}} A^2 \quad (16)$$

so

$$A^{20} = 3^9 \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \quad (17)$$

4. [20 pts] Determinant Using Levi-Civita Symbol

The determinant of an arbitrary 3×3 matrix A can be defined as

$$\det A = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \epsilon_{ijk} A_{1i} A_{2j} A_{3k},$$

where ϵ_{ijk} is completely antisymmetric and $\epsilon_{123} = +1$.

(a) Using the above definition, show that interchanging column 1 and column 3 of A changes the sign of $\det A$.

(b) Evaluate

$$\int_0^{\pi/2} \begin{vmatrix} 1 & \cos x & 1 - \cos x \\ 1 + \sin x & \cos x & 1 + \sin x - \cos x \\ \sin x & \sin x & 1 \end{vmatrix} dx.$$

(c) Consider

$$B = \begin{pmatrix} (1-x)^{22} & (1-x)^{44} & (1-x)^{66} \\ (1-x)^{33} & (1-x)^{66} & (1-x)^{99} \\ (1-x)^{44} & (1-x)^{88} & (1-x)^{1212} \end{pmatrix}.$$

Find the coefficient of x^1 in $\det B$.

Answer:

(a) The original A 's determinant is

$$\det A = \sum_i \sum_k \sum_j \epsilon_{ijk} a_{1i} a_{2j} a_{3k} \quad (18)$$

The determinant of changed A is

$$\det A' = \sum_i \sum_k \sum_j \epsilon_{ijk} a_{3i} a_{2j} a_{1k} = \sum_i \sum_k \sum_j \epsilon_{kji} a_{1i} a_{2j} a_{3k} \quad (19)$$

For all i, j, k ,

$$\epsilon_{ijk} = -\epsilon_{ikj} = \epsilon_{kij} = -\epsilon_{kji} \quad (20)$$

As a result,

$$\det A' = \sum_i \sum_k \sum_j \epsilon_{kji} a_{1i} a_{2j} a_{3k} = - \sum_i \sum_k \sum_j \epsilon_{ijk} a_{1i} a_{2j} a_{3k} = -\det A \quad (21)$$



(b) The function can be derived:

$$f(x) = \begin{vmatrix} 1 & \cos x & 1 - \cos x \\ \sin x & 0 & \sin x \\ \sin x & \sin x & 1 \end{vmatrix} = -\sin x \cos x \quad (22)$$

As a result,

$$\int_0^{\frac{\pi}{2}} \sin x \cos x dx = -\frac{1}{2} \quad (23)$$

(c) The determinant of B is

$$\det B = \sum_{i,j,k} \epsilon_{ijk} b_{1i} b_{2j} b_{3k} \quad (24)$$

Every terms contribution to the coefficient of x^1 are listed as follow:

$$\begin{aligned} (i, j, k) &= (1, 2, 3) = -1 \cdot (22 + 66 + 1212) \\ (i, j, k) &= (1, 3, 2) = -(-1)(22 + 99 + 88) \\ (i, j, k) &= (2, 1, 3) = -(-1)(44 + 33 + 1212) \\ (i, j, k) &= (2, 3, 1) = -1 \cdot (44 + 99 + 44) \\ (i, j, k) &= (3, 1, 2) = -1 \cdot (66 + 33 + 88) \\ (i, j, k) &= (3, 2, 1) = -(-1)(66 + 66 + 44) \end{aligned}$$

and the total is 0

5. [25 pts] Pauli Spin Matrices

In quantum mechanics, the three Pauli spin matrices S_1, S_2, S_3 , which are Hermitian, and the 2×2 unit matrix I are:

$$S_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The products of Pauli matrices satisfy:

$$S_m S_n = \delta_{mn} I + i \sum_{k=1}^3 \epsilon_{mnk} S_k, \quad m, n, k = 1, 2, 3.$$

(a) Define the commutator and anti-commutator:

$$[A, B] = AB - BA, \quad \{A, B\} = AB + BA.$$

Evaluate $[S_m, S_n]$ and $\{S_m, S_n\}$.

(b) Given a vector $\mathbf{v} = (v_1, v_2, v_3)$, define

$$S(\mathbf{v}) = \sum_{i=1}^3 v_i S_i = v_1 S_1 + v_2 S_2 + v_3 S_3.$$

Prove that

$$S(\mathbf{a})S(\mathbf{b}) = (\mathbf{a} \cdot \mathbf{b})I + iS(\mathbf{a} \times \mathbf{b}),$$

and deduce a condition for $\mathbf{a}$ and $\mathbf{b}$ so that $S(\mathbf{a})$ and $S(\mathbf{b})$ commute.



(c) For a unit vector $\mathbf{n}$ and real θ , show that

$$\exp[i\theta S(\mathbf{n})] = \cos \theta I + i \sin \theta S(\mathbf{n}).$$

Answer:

(a)

$$[S_m, S_n] = \delta_{nm}I + i \sum_k \epsilon_{mnk} S_k - \delta_{mn}I - i \sum_k \epsilon_{nmk} S_k = 2i \sum_k \epsilon_{nmk} S_k \quad (25)$$

$$\{S_m, S_n\} = \delta_{nm}I + i \sum_k \epsilon_{mnk} S_k + \delta_{mn}I + i \sum_k \epsilon_{nmk} S_k = 2\delta_{mn}I \quad (26)$$

(b)

$$\begin{aligned} S(\mathbf{a})S(\mathbf{b}) &= \left(\sum_i a_i S_i \right) \left(\sum_j b_j S_j \right) = \sum_{i,j} a_i b_j S_i S_j \\ &= \sum_{i,j} a_i b_j (\delta_{ij}I + i \sum_k \epsilon_{ijk} S_k) = \sum_{i,j} \delta_{ij} a_i b_j I + \sum_i \sum_{j,k} \epsilon_{ijk} a_i b_j S_k \\ &= (\mathbf{a} \cdot \mathbf{b})I + \sum_i (\mathbf{a} \times \mathbf{b})_i S_i = (\mathbf{a} \cdot \mathbf{b})I + iS(\mathbf{a} \times \mathbf{b}) \end{aligned} \quad (27)$$

When $S(\mathbf{a})$ and $S(\mathbf{b})$ commute

$$[S(\mathbf{a}), S(\mathbf{b})] = 2iS(\mathbf{a} \times \mathbf{b}) = 0 \quad (28)$$

which implies

$$\mathbf{a} \times \mathbf{b} = 0 \quad (29)$$

vector $\mathbf{a}$ and $\mathbf{b}$ is vertical.

(c) It is obviously that

$$S^2(\mathbf{n}) = |\mathbf{n}|^2 I + iS(\mathbf{n} \times \mathbf{n}) = I, S^3(\mathbf{n}) = S(\mathbf{n}) \quad (30)$$

As a result,

$$\begin{aligned} \exp[i\theta S(\mathbf{n})] &= \sum_{r=0}^{\infty} \frac{[i\theta S(\mathbf{n})]^{2r}}{(2r)!} + \sum_{r=1}^{\infty} \frac{[i\theta S(\mathbf{n})]^{2r-1}}{(2r-1)!} \\ &= \sum_{r=0}^{\infty} (-1)^r \frac{\theta^{2r}}{(2r)!} I + i \sum_{r=1}^{\infty} (-1)^{r-1} \frac{\theta^{2r-1}}{(2r-1)!} S(\mathbf{n}) \\ &= \cos \theta I + i \sin \theta S(\mathbf{n}) \end{aligned} \quad (31)$$

Q.E.D.

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